

```

*****
*                                     *
*          STAAD.Pro                 *
*          Version  2005      Bld 1002.US      *
*          Proprietary Program of           *
*          Research Engineers, Intl.         *
*          Date=    AUG 18, 2006            *
*          Time=    12:24:44                *
*                                     *
*          USER ID: Meinhardt Singapore Pte Ltd      *
*****

```

```

1. STAAD PLANE FRAME 2
INPUT FILE: JA_BLD 1_FR2.STD
2. START JOB INFORMATION
3. JOB NAME - DUBAI METRO PROJECT
4. JOB CLIENT - GOVERNMENT OF DUBAI _ ROADS _TRANSPORT AUTHORITY
5. ENGINEER DATE 17-AUG-06
6. END JOB INFORMATION
7. INPUT WIDTH 79
8. UNIT METER KN
9. JOINT COORDINATES
10. 1 0 6.35 0; 2 1.5 6.35 0; 3 26.6 6.35 0; 4 28.1 6.35 0; 11 28.1 3.55 0
11. 12 29.6 3.55 0; 13 32.1 3.55 0; 14 33.6 3.55 0; 101 0 0 0; 201 28.1 0 0
12. 301 33.6 0 0
13. MEMBER INCIDENCES
14. 1 1 2; 2 2 3; 3 3 4; 11 11 12; 12 12 13; 13 13 14; 101 101 1; 201 201 11
15. 202 11 4; 301 301 14
16. MEMBER PROPERTY BRITISH
17. 301 TABLE ST UC254X254X73
18. MEMBER PROPERTY AMERICAN
19. 1 TO 3 101 201 202 TABLE ST W30X124
20. 11 TO 13 TABLE ST W12X26
21. DEFINE MATERIAL START
22. ISOTROPIC STEEL
23. E 2.05E+008
24. POISSON 0.3
25. DENSITY 76.8195
26. ALPHA 1.2E-005
27. DAMP 0.03
28. END DEFINE MATERIAL
29. CONSTANTS
30. MATERIAL STEEL MEMB 1 TO 3 11 TO 13 101 201 202 301
31. MEMBER RELEASE
32. 11 START MZ
33. SUPPORTS
34. 101 201 301 PINNED
35. LOAD 1 SELFWEIGHT
36. SELFWEIGHT Y -1
37. LOAD 2 SDL
38. JOINT LOAD
39. 1 FY -32.5
40. 4 FY -17.5

```

FRAME 2

--- PAGE NO. 2

41. 14 FY -20.5
42. MEMBER LOAD
43. 1 TO 3 11 TO 13 UNI GY -6.4
44. LOAD 3 LL
45. JOINT LOAD
46. 4 14 FY -3.4
47. MEMBER LOAD
48. 1 TO 3 11 TO 13 UNI GY -4.8
49. 2 CON GY -10 12.55
50. 2 CON GY -8 1.4
51. 2 CMOM GZ -2 1.4
52. 2 CON GY -8 7.4
53. 2 CMOM GZ -2 7.4
54. 2 CON GY -8 14.6
55. 2 CMOM GZ -2 14.6
56. 2 CON GY -8 20.6
57. 2 CMOM GZ -2 20.6
58. LOAD 4 PH (NOTIONAL HORIZONTAL LOAD)
59. JOINT LOAD
60. 1 FX 4.2
61. LOAD 11 WL1 (INTERNAL SUCTION, WIND ANGLE 0)
62. MEMBER LOAD
63. 101 UNI GX 9
64. 202 301 UNI GX 2
65. 1 UNI GY -1.3
66. 2 UNI GY -1.3 0 12.55
67. 2 UNI GY -5 12.55 25.1
68. 3 UNI Y -5
69. 11 TO 13 UNI GY -2
70. LOAD 12 WL2 (INTERNAL PRESSURE, WIND ANGLE 0)
71. MEMBER LOAD
72. 101 UNI GX 4
73. 202 301 UNI GX 6.9
74. 1 UNI GY 6.2
75. 2 UNI GY 6.2 0 12.55
76. 2 UNI GY 4 12.55 25.1
77. 3 UNI Y 4
78. 11 TO 13 UNI GY -5.3
79. LOAD 13 WL3 (INTERNAL SUCTION, WIND ANGLE 90)
80. MEMBER LOAD
81. 101 UNI GX -5
82. 202 301 UNI GX 5
83. 1 TO 3 11 TO 13 UNI GY -5
84. LOAD 14 WL4 (INTERNAL PRESSURE, WIND ANGLE 90)
85. MEMBER LOAD
86. 101 UNI GX -9.9
87. 202 301 UNI GX 9.9
88. 1 TO 3 11 TO 13 UNI GY 8.9
89. LOAD 15 WL5 (INTERNAL SUCTION, WIND ANGLE 0, REVERSE WIND)
90. MEMBER LOAD
91. 101 UNI GX -2
92. 202 301 UNI GX -8.9
93. 1 UNI GY -2.1
94. 2 UNI GY -2.1 0 12.55
95. 2 UNI GY -5 12.55 25.1
96. 3 UNI GY -5

FRAME 2

-- PAGE NO. 3

97. 11 TO 13 UNI GY -1.7
98. LOAD 16 WL6 (INTERNAL PRESSURE, WIND ANGLE 0, REVERSE WIND)
99. MEMBER LOAD
100. 101 UNI GX -6.9
101. 202 301 UNI GX -4
102. 1 UNI GY 5.8
103. 2 UNI GY 5.8 0 12.55
104. 2 UNI GY 4 12.55 25.1
105. 3 UNI Y 4
106. 11 TO 13 UNI GY 3.3
107. LOAD 21 EQ (EARTHQUAKE LOAD)
108. MEMBER LOAD
109. 101 UNI GX 11
110. 201 202 UNI GX 11.2
111. 301 UNI GX 5.3
112. LOAD COMB 1001 1.4DL + 1.6LL
113. 1 1.4 2 1.4 3 1.6
114. LOAD COMB 1002 1.4DL + 1.6LL + 1.0PH
115. 1 1.4 2 1.4 3 1.6 4 1.0
116. LOAD COMB 1011 1.2DL + 1.2LL + 1.2WL1
117. 1 1.2 2 1.2 3 1.2 11 1.2
118. LOAD COMB 1111 1.4DL + 1.4WL1
119. 1 1.4 2 1.4 11 1.4
120. LOAD COMB 1211 1.0DL + 1.4WL1
121. 1 1.0 2 1.0 11 1.4
122. LOAD COMB 1012 1.2DL + 1.2LL + 1.2WL2
123. 1 1.2 2 1.2 3 1.2 12 1.2
124. LOAD COMB 1112 1.4DL + 1.4WL2
125. 1 1.4 2 1.4 12 1.4
126. LOAD COMB 1212 1.0DL + 1.4WL2
127. 1 1.0 2 1.0 12 1.4
128. LOAD COMB 1013 1.2DL + 1.2LL + 1.2WL3
129. 1 1.2 2 1.2 3 1.2 13 1.2
130. LOAD COMB 1113 1.4DL + 1.4WL3
131. 1 1.4 2 1.4 13 1.4
132. LOAD COMB 1213 1.0DL + 1.4WL3
133. 1 1.0 2 1.0 13 1.4
134. LOAD COMB 1014 1.2DL + 1.2LL + 1.2WL4
135. 1 1.2 2 1.2 3 1.2 14 1.2
136. LOAD COMB 1114 1.4DL + 1.4WL4
137. 1 1.4 2 1.4 14 1.4
138. LOAD COMB 1214 1.0DL + 1.4WL4
139. 1 1.0 2 1.0 14 1.4
140. LOAD COMB 1015 1.2DL + 1.2LL + 1.2WL5
141. 1 1.2 2 1.2 3 1.2 15 1.2
142. LOAD COMB 1115 1.4DL + 1.4WL5
143. 1 1.4 2 1.4 15 1.4
144. LOAD COMB 1215 1.0DL + 1.4WL5
145. 1 1.0 2 1.0 15 1.4
146. LOAD COMB 1016 1.2DL + 1.2LL + 1.2WL6
147. 1 1.2 2 1.2 3 1.2 16 1.2
148. LOAD COMB 1116 1.4DL + 1.4WL6
149. 1 1.4 2 1.4 16 1.4
150. LOAD COMB 1216 1.0DL + 1.4WL6
151. 1 1.0 2 1.0 16 1.4
152. LOAD COMB 1021 1.2DL + 0.5LL + 1.0EQ

FRAME 2

-- PAGE NO. 4

```

153. 1 1.2 2 1.2 3 0.5 21 1.0
154. LOAD COMB 1121 1.2DL + 0.5LL - 1.0EQ
155. 1 1.2 2 1.2 3 0.5 21 -1.0
156. LOAD COMB 1221 0.9DL + 1.0EQ
157. 1 0.9 2 0.9 21 1.0
158. LOAD COMB 1321 0.9DL - 1.0EQ
159. 1 0.9 2 0.9 21 -1.0
160. LOAD COMB 2001 1.0DL + 1.0LL
161. 1 1.0 2 1.0 3 1.0
162. LOAD COMB 2002 1.0DL + 0.5LL
163. 1 1.0 2 1.0 3 0.5
164. LOAD COMB 2011 DL + LL + WL1
165. 1 1.0 2 1.0 3 1.0 11 1.0
166. LOAD COMB 2012 DL + LL + WL2
167. 1 1.0 2 1.0 3 1.0 12 1.0
168. LOAD COMB 2013 DL + LL + WL3
169. 1 1.0 2 1.0 3 1.0 13 1.0
170. LOAD COMB 2014 DL + LL + WL4
171. 1 1.0 2 1.0 3 1.0 14 1.0
172. LOAD COMB 2015 DL + LL + WL5
173. 1 1.0 2 1.0 3 1.0 15 1.0
174. LOAD COMB 2016 DL + LL + WL6
175. 1 1.0 2 1.0 3 1.0 16 1.0
176. PERFORM ANALYSIS

```

PROBLEM STATISTICS

```

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 11/ 10/ 3
ORIGINAL/FINAL BAND-WIDTH= 8/ 2/ 7 DOF
TOTAL PRIMARY LOAD CASES = 11, TOTAL DEGREES OF FREEDOM = 27
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 855.1 MB

```

```

177. PARAMETER
178. CODE BS5950
179. UNL 3 MEMB 101
180. LZ 28 MEMB 101
181. LY 3 MEMB 101
182. UNL 3.55 MEMB 201 202
183. LY 3.55 MEMB 201 202
184. UNL 3.55 MEMB 301
185. LZ 10 MEMB 301
186. LY 3 MEMB 301
187. UNL 3 MEMB 1 TO 3
188. LZ 28 MEMB 1 TO 3
189. LY 1.5 MEMB 1 TO 3
190. UNL 1.5 MEMB 11 12
191. UNL 3 MEMB 13
192. LZ 21 MEMB 11 TO 13

```

FRAME 2

-- PAGE NO. 5

193. LY 1.5 MEMB 11 TO 13
194. *TRACK 2.0 MEMB 101
195. CHECK CODE ALL

STAAD.Pro CODE CHECKING - (BSI)

PROGRAM CODE REVISION V2.9_5950-1_2000